Chapter 6: Velocity Fields

Reading: [Fung/94] Chapter 6

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2026-05-06

Contents

(§6.1) Velocity Fields

Fluids

- In the study if a fluid, we are generally concerned with the a flow, and it's velocity field. The field of flow is described by the time rate of change in the position vector $\dot{\boldsymbol{x}}$, called the velocity vector field.
- The velocity components in indicial notation read, $v_i = v_i(x_1,\, x_2,\, x_3, t)$.
- The velocity difference between two adjacent points P and P' located at x_i and $x_i + dx_i$ can be found by invoking the chain rule of differentiation as
$$dv_i = \frac{\partial v_i}{\partial x_j} dx_j.$$
- This also corresponds to the operation
$$d\boldsymbol{v} = d\boldsymbol{x} \cdot \nabla \boldsymbol{v}$$
where $\nabla \boldsymbol{v}$ is known as the *velocity gradient* tensor.

Cartan Decomposition of the Velocity Gradient

The *Cartan* decomposition of the velocity gradient vector yields

$$\frac{\partial v_i}{\partial x_j} = \frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right) + \frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} - \frac{\partial v_j}{\partial x_i}\right)$$

where the left term on the right hand side is a symmetric tensor and the one on the right is a skew-symmetric tensor.

Definitions

- The symmetric part of the velocity gradient is the rate of deformation tensor, (or the deformation rate tensor)

$$V_{ij} \equiv \frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right),$$

- while the skew-symmetric part is called the spin tensor

$$\Omega_{ij} \equiv \frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} - \frac{\partial v_j}{\partial x_i}\right).$$

- Then

$$\frac{\partial v_i}{\partial x_j} = V_{ij} + \Omega_{ij}$$

- Evidently, V_{ij} is symmetric and Ω_{ij} is skew-symmetric

$$V_{ij} = V_{ji}, \qquad \Omega_{ij} = -\Omega_{ji}.$$

Vorticity

- The tensor Ω_{ij} has only three independent (non-zero) elements

$$[\Omega_{ij}] = \begin{bmatrix} 0 & \Omega_{12} & \Omega_{13} \\ -\Omega_{12} & 0 & \Omega_{23} \\ -\Omega_{13} & -\Omega_{23} & 0 \end{bmatrix}$$

- There exists a vector $\boldsymbol{\omega}$ dual to $\boldsymbol{\Omega}$; that is

$$\omega_k \equiv \varepsilon_{kij}\Omega_{ij}; \qquad \text{i.e.} \qquad \boldsymbol{\omega} = \nabla \times \boldsymbol{v} = \operatorname{curl} \boldsymbol{v}$$

where ε_{kij} is the permutation tensor, and the vector $\boldsymbol{\omega}$ is called the *vorticity* vector.

Query AI:

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Provide a clear definition of vorticity in a flow field.
If possible, provide illustrations or suggest references or sources
that point graphical illustrations depicting vorticity.
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Problems

6.1

6.1 Consider the motion of a body fluid with velocity components u and v derived from a potential Φ

$$u = \frac{\partial \Phi}{\partial x}, \qquad v = \frac{\partial \Phi}{\partial y},$$

while the component w is identically zero. Sketch the velocity field for the following potentials:

(a) $\Phi = \dfrac{1}{4\pi} \log (x^2 + y^2) = \dfrac{1}{2\pi} \log r, \qquad (r^2 = x^2 + y^2)$

(b) $\Phi = x$

(c) $\Phi = Ar^n \cos n\theta, \qquad \left(\theta = \tan^{-1} \dfrac{y}{x}\right)$

(d) $\Phi = \dfrac{\cos \theta}{r}$

Acceleration

Acceleration is the time rate of change of velocity.

Derivation Process

- As the velocity field has been provided in terms of the current (moving) coordinate system, i.e., x_1, x_2, x_3, its derivate needs to be handled with care.
- Remember the derivative of a function of function, i.e. $G(\boldsymbol{x}(t), t)$

$$\frac{DG}{Dt} = \dot{G} = \frac{\partial G}{\partial t} + \frac{\partial \boldsymbol{x}}{\partial t} \cdot \frac{\partial G}{\partial \boldsymbol{x}} \tag{1}$$

$$= \frac{\partial G}{\partial t} + \dot{\boldsymbol{x}} \cdot \nabla G \tag{2}$$

- Replacing G with $\boldsymbol{v} = \dot{\boldsymbol{x}}$, we get

$$\dot{\boldsymbol{v}} = \frac{\partial \boldsymbol{v}}{\partial t} + \boldsymbol{v} \cdot \nabla \boldsymbol{v}$$

- The second term on the right hand side is known as the *convective* (or sometimes as the *advective*) derivative term, and it appears when a field is represented in terms of the moving coordinates.

Exercise

Express the acceleration vector in indicial notation.